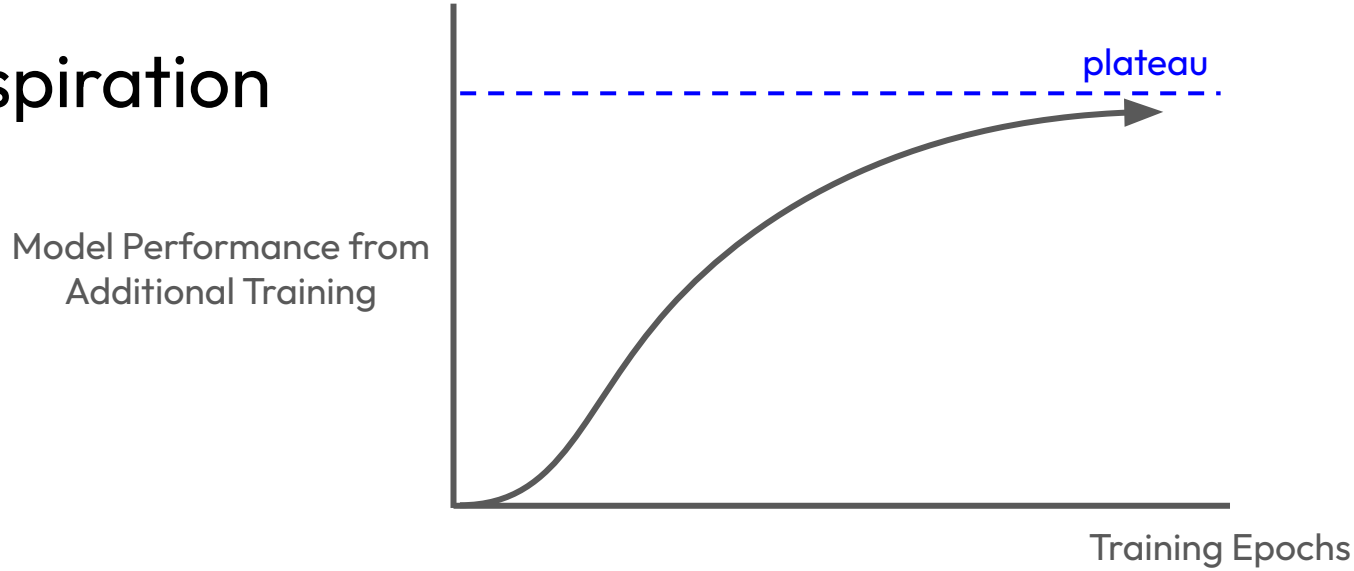
MMRC: Multimodal Reasoning and Critique for Web Navigation

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UC Berkeley: Prof. Pieter Abbeel and Prof. Ion Stoica
Google DeepMind: Izzeddin Gur, Hiroki Furuta, and Aleksandra Faust

Inspiration



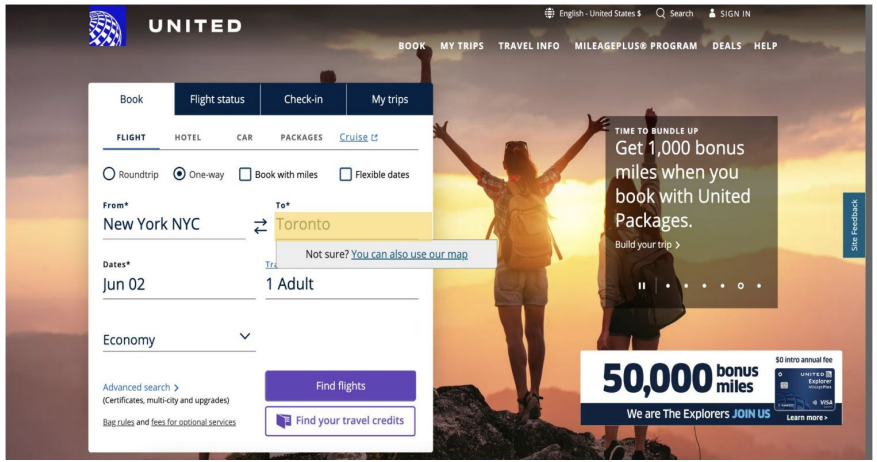
1. Use data smarter

2. Higher order logic

MCTS, **actor-critic**, CoT, transfer learning, etc.

How can we apply actor-critic to boost the performance of automated web navigation?

Automated Web Navigation



Task: “Find one-way flights from New York to Toronto.” (Source: Xiang Deng et al.)

HTML, webpage image, previous actions (H, I, M)

↓

Output: Agent action (e, ϕ, v)

e: webpage element to act on
ϕ: operation (click, type select)
v: additional values (e.g. “NYC”)

Can we improve the performance of out-of-the-box models with little to no finetuning?

Contributions

Jai
★ Setting up training environment
★ Implementing actor-critic interaction

Abdus
★ Guidance on project direction
★ Finetuning small LM critic model
★ Running experiments and reporting results

Professors provided mentorship and guidance.

Related Works

MindAct
- ★ Small LM to filter 50 HTML elements
- ★ Generates a series of MCQs to find correct element

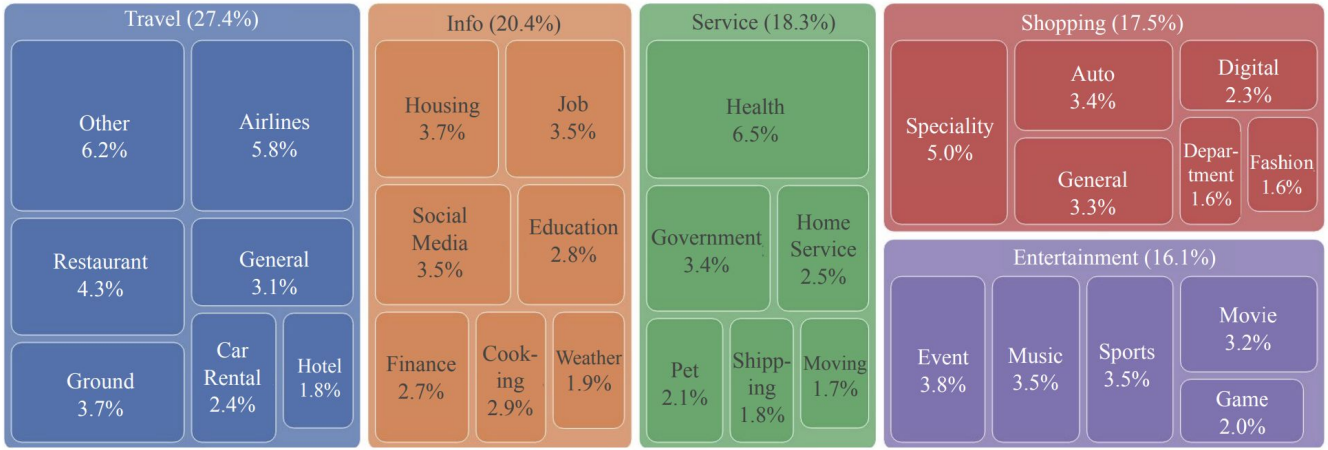
SeeAct
- ★ MindAct with visual input
- ★ Image annotations + multi round textual MCQs
- ★ Manual action grounding

“A Real-World WebAgent…”
- ★ HTML-T5 model for long HTML documents
- ★ Summarizes long HTML documents into task-relevant snippets

MMRC
- ★ Only one actor query
- ★ No multiple rounds
- ★ Visual actor-critic

Dataset

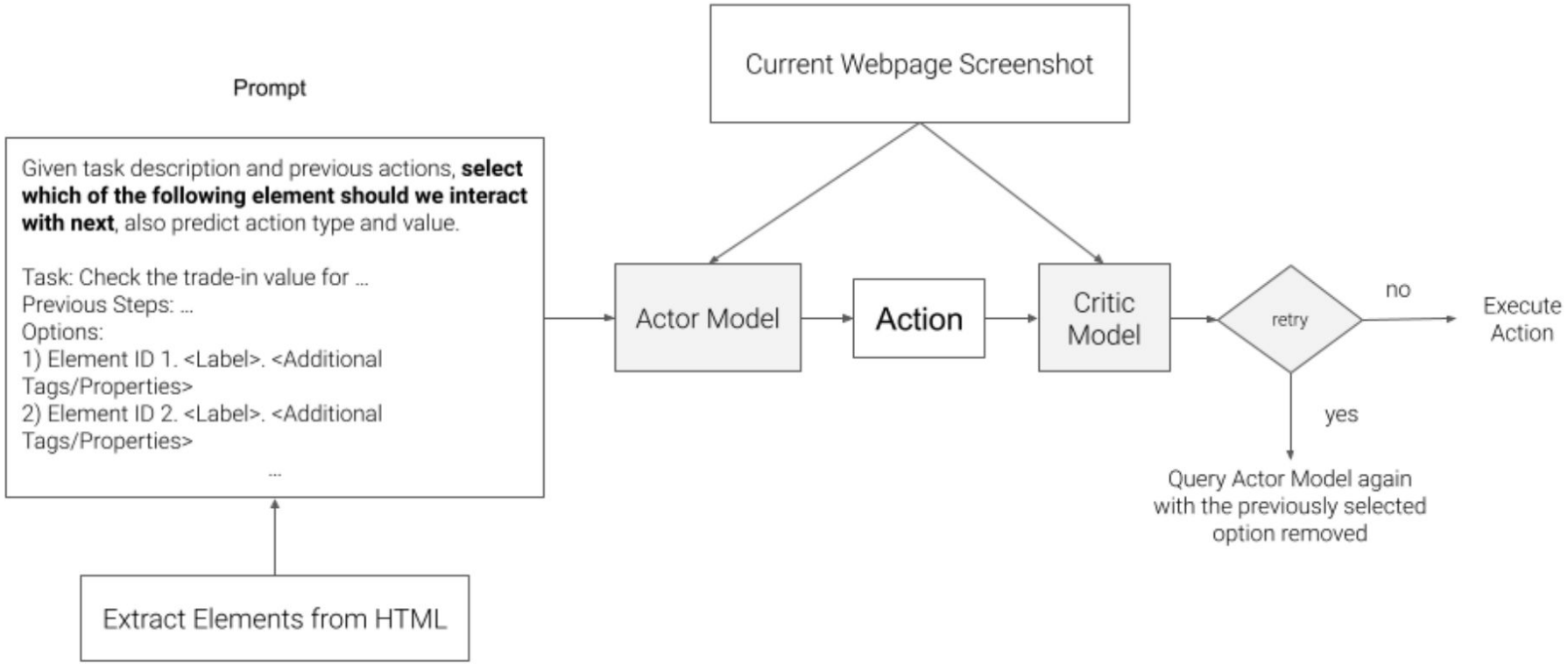
Multimodal Mind2Web



Source: Xiang Deng et al.

137 real world websites
31 domains
Cross Task
Cross Website
Cross Domain
Authentic noisiness

MMRC Actor Critic Setup



MMRC Setup

Algorithm 2 MMRC

Input: Website $\mathcal{W}$, task $\mathcal{T}$, and maximum critic steps N
$t \leftarrow 0$
$M \leftarrow \emptyset$
while $TERMINATION_CONDITION$ is not met **do**
  $\mathcal{O} \leftarrow$ Set of textual representations of the elements of the current webpage HTML H
  $\mathcal{O} \leftarrow FILTER(\mathcal{O})$
  $\mathcal{P} \leftarrow \emptyset$
  for $i \leftarrow 1$ to N **do**
    $a \leftarrow$ Predict next action with π_{Actor} using prompt template listed in Listing 4.1
    $\mathcal{P} \leftarrow \mathcal{P} \cup \{a\}$
    $d \leftarrow$ Critique the predicted action a using π_{Critic}
    if d is True **then**
      break
    end if
    $\mathcal{O} \leftarrow \mathcal{O} - \{a\}$
  end for
  $a_{sel} \leftarrow SELECT_ACTION(\mathcal{P})$
  Execute a_{sel} on the webpage w
  $M \leftarrow M \cup \{a_{sel}\}$
  $t \leftarrow t + 1$
end while

How is MMRC Different?

- Actor critic with visual input
- Small finetuned critic model
- MCQ in one shot

Listing 4.2: MMRC Critic Prompt

```
{The webpage screenshot}
You are an automated web navigation evaluator. Given a task, a webpage screenshot, previous navigation steps, and a suggested next step: decide whether we should execute the suggested next step or, consider a different step to complete the task.
Task: {The Task Description}
Previous Steps:
{History of the previous actions taken}
Next Step Suggestion:
{Precited Action}
Based on the previous steps taken and the current state of the website, suggest whether we should execute the suggested next step to complete the task ({The Task Description}).
Structure your suggestion strictly in the following format:
decision: yes/no
explanation: Brief sentence explaining why we should execute/not execute the suggested next step.
```

screenshot, task, history, predicted action

Listing 4.1: MMRC Actor Prompt

```
{The webpage screenshot}
You are an automated web navigation assistant: Given a task, a webpage screenshot, previous steps, and a list of elements from the webpage, you need to suggest which element from the provided list to act upon in the current step.
Task: {The Task Description}
Previous Steps: {History of the previous actions taken}
Current Step Options:
You must act upon exactly one of the following webpage elements, where each element is formatted as:
ID. [element type] label <Additional properties separated by semicolon>(optional).
{Enumerated List of Interactive Elements, represented with short textual description}
Based on the screenshot, previous steps, and the provided options, summarize how much progress has already been made and suggest which element from the list to act upon in the current step. An action can be one of [CLICK, TYPE, SELECT], where SELECT means selecting an option from dropdown/radio buttons.
Structure your suggestion strictly in the following format:
progress_made: Brief sentence explaining the current progress towards completing {task_description}.
next_step: Suggest the next step based on the provided list of elements with brief explanation/rationale
element_id: Exact ID of the selected element from the provided list.
action: CLICK/TYPE/SELECT
value: In case of TYPE, value is the text we should type. For SELECT, value corresponds to the option we are selecting. For CLICK value is "None".
```

screenshot, HTML, task, history

List of Hypotheses to Test

- ★ H1: Can MMRC improve web agents?
- ★ H2: Can a smaller finetuned critic perform better than a foundational critic model?
- ★ H3: MMRC vs other multimodal methods
- ★ H4: Number of iterations vs performance

Evaluation

Element Accuracy

selected element vs acceptable options

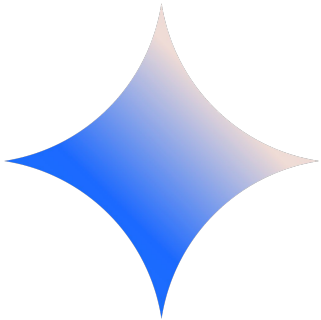
Operation F1

token level F1 score for operation prediction

Step Success Rate

element and operation are correct

Models Used



Gemini 1.5 Pro (actor, critic)
Gemini 1.0 Pro Vision (actor, critic)



Phi-3 Model (critic)

Finetuned on only textual input:
Batch 64, LoRA fp16, Rank 128, alpha 256, LR 2e-4

Results for H1

Base Actor	Method Name	Cross Task		Cross Website		Cross Domain	
		Step SR	Relative Improvement (%)	Step SR	Relative Improvement (%)	Step SR	Relative Improvement (%)
Gemini 1.5 Pro	Baseline - No Critic	28.83		26.89		28.87	
	MMRC with Phi-3 Vision Critic	29.72	3.09	27.38	1.82	30	3.91
	MMRC with Gemini 1.5 Pro Critic	30.69	6.45	28.56	6.21	30.27	**4.85**
Gemini 1.0 Pro Vision	Baseline - No Critic	19.05		15.44		21.37	
	MMRC with Gemini 1.0 Pro Vision Critic	20.49	**7.56**	17.19	**11.33**	22.35	4.59

MMRC is better than base actor, so it can improve web agents.

Results for H2

Base Actor	Method Name	Cross Task			Cross Website			Cross Domain		
		Element Accuracy	Operation F1	Step SR	Element Accuracy	Operation F1	Step SR	Element Accuracy	Operation F1	Step SR
Gemini 1.5 Pro	Baseline - No Critic	38.54	71.65	28.83	37.1	71.78	26.89	36.63	74.5	28.87
	MMRC with Phi-3 Critic	39.36	71.7	29.72	33.1	**75.9**	27.38	38.2	**74.3**	30
	MMRC with Gemini 1.5 Pro Critic	**39.88**	**72.15**	**30.69**	**39.55**	72.75	**28.56**	**38.57**	73.7	**30.27**
Gemini 1.0 Pro Vision	Baseline - No Critic	22.3	72.04	19.05	18.79	70.26	15.44	24.7	73.83	21.37
	MMRC with Gemini 1.0 Pro Vision Critic	23.59	72.77	20.49	20.39	71.21	17.19	25.58	74.08	22.35

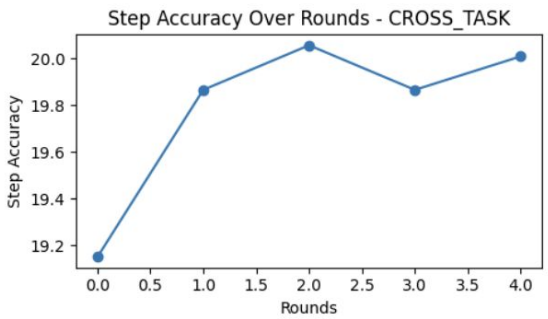
Smaller critics have the potential to perform better, but need to be finetuned on visual data too.

Results for H3

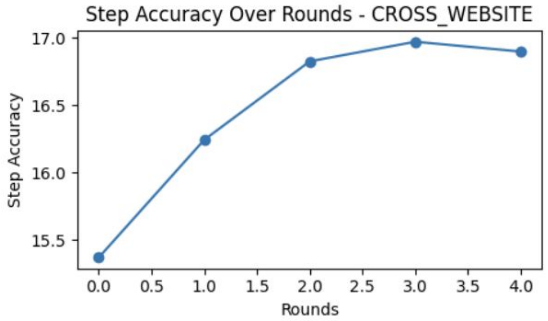
Method Name	Cross Task			Cross Website			Cross Domain		
	Element Accuracy	Operation F1	Step SR	Element Accuracy	Operation F1	Step SR	Element Accuracy	Operation F1	Step SR
SeeAct - with Gemini 1.0 Pro Vision - MCQ	21.5	67.7	19.6	17.1	61.3	15.4	20.7	64.3	18
SeeAct - with GPT4 + MCQ	**46.4**	**73.8**	**40.2**	38	67.8	**32.4**	**42.4**	69.3	**36.8**
SeeAct - with GPT4 + Human Oracle	66.4	79.2	61.9	69.5	78.9	65	72.8	73.6	62.1
MMRC with Gemini 1.0 Pro Vision	23.59	72.77	20.49	20.39	71.21	17.19	25.58	74.08	22.35
MMRC with Gemini Pro 1.5 Actor and Phi-3 Vision Critic	39.36	71.7	29.72	33.1	**75.9**	27.38	38.2	**74.3**	30
MMRC with Gemini Pro 1.5	39.88	72.15	30.69	**39.55**	72.75	28.56	38.57	73.7	30.27

GPT-4V is better in most cases, but it was grounded manually by human supervision.
MMRC outperforms other methods in several cases, validating the small critic approach.

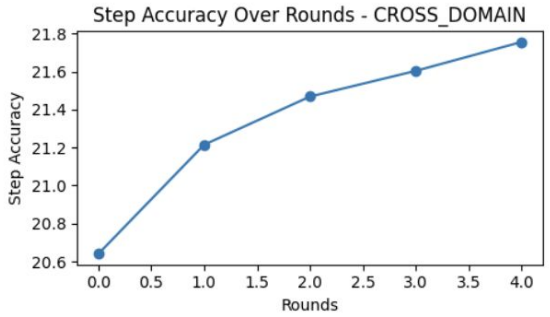
Results for H4



(a) Cross Task



(b) Cross Website



(c) Cross Domain

In initial actor-critic rounds, the step accuracy improves dramatically. Then, it starts to plateau.

Future Work

1. Finetune small critics on both visual and textual input
2. Discriminate against lacking HTML descriptions (e.g. <img>)

Citations

Xiang Deng et al. “Mind2web: Towards a generalist agent for the web”. In: Advances in Neural Information Processing Systems 36 (2024).

Boyuan Zheng et al. “Gpt-4v (ision) is a generalist web agent, if grounded”. In: arXiv preprint arXiv:2401.01614 (2024).

Izzeddin Gur et al. A Real-World WebAgent with Planning, Long Context Understanding, and Program Synthesis. 2024. arXiv: 2307.12856 [cs.LG]. URL: https://arxiv.org/abs/2307.12856.

Link to paper (has more citations)